TCP congestion control comparison

Completed matched destinations: 3/3. Requested algorithms: cubic, reno, bbr. Repetitions per algorithm: 3. Duration: 60 s. Sampling: 1.0 s. Seed: 422. Host kernel: 6.18.33.2-microsoft-standard-WSL2.

Test order: seeded initial shuffle; rotate by repetition and candidate index. Each test and its status are recorded in manifest.json. Destinations were shuffled once, then retained only when every algorithm and repetition succeeded. The first complete destination is the representative, chosen without comparing throughput. Failed candidates remain in the manifest and successful partial measurements remain on disk.

The per-run summary gives minimum, median, arithmetic mean, and P95 sampled goodput, duration-weighted goodput, mean/P95 RTT, and retransmitted segments. P95 uses linear interpolation at index (n - 1) * 0.95 in the sorted samples. Aggregate throughput is the mean of duration-weighted per-run goodputs; variability is sample standard deviation across runs. RTT aggregates average the sampled smoothed RTT, not independent packet RTT observations.

dal.speedtest.clouvider.net:5200: highest measured mean goodput: reno (281.301 Mbit/s). Lowest measured mean RTT: bbr (64.530 ms).

- bbr: mean goodput 269.791 Mbit/s, run standard deviation 11.688, range 256.316-277.171.
- cubic: mean goodput 273.962 Mbit/s, run standard deviation 5.633, range 268.468-279.725.
- reno: mean goodput 281.301 Mbit/s, run standard deviation 2.756, range 279.145-284.405.

speed.netfiber.net.il:5201: highest measured mean goodput: reno (119.066 Mbit/s). Lowest measured mean RTT: bbr (177.386 ms).

- reno: mean goodput 119.066 Mbit/s, run standard deviation 0.893, range 118.039-119.667.
- bbr: mean goodput 106.588 Mbit/s, run standard deviation 17.365, range 86.572-117.631.
- cubic: mean goodput 118.085 Mbit/s, run standard deviation 4.160, range 113.306-120.890.

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96.45.41.167:30001: highest measured mean goodput: reno (163.523 Mbit/s). Lowest measured mean RTT: cubic (132.581 ms).

  cubic: mean goodput 163.117 Mbit/s, run standard deviation 17.505, range 143.268-176.351.

  reno: mean goodput 163.523 Mbit/s, run standard deviation 11.802, range 152.883-176.217.

  bbr: mean goodput 147.472 Mbit/s, run standard deviation 6.018, range 141.150-153.131.

Per-run statistics (goodput in Mbit/s; RTT in ms):

dal.speedtest.clouvider.net:5200 bbr run 1: goodput min/median/mean/P95 = 202.814 / 260.731 / 256.315 / 281.472; mean RTT = 62.459; retransmitted segments = 104; matched destination = True.

Window/RTT context: minimum filtered RTT 28.965 ms; median cwnd 2636.0 segments (3816928 bytes); observed-rate BDP proxy 928023 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 cubic run 1: goodput min/median/mean/P95 = 208.026 / 275.568 / 273.692 / 287.345; mean RTT = 66.713; retransmitted segments = 121; matched destination = True.

Window/RTT context: minimum filtered RTT 30.309 ms; median cwnd 1886.0 segments (2730928 bytes); observed-rate BDP proxy 1036914 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 reno run 1: goodput min/median/mean/P95 = 219.239 / 281.231 / 279.146 / 293.892; mean RTT = 73.369; retransmitted segments = 345; matched destination = True.

Window/RTT context: minimum filtered RTT 31.328 ms; median cwnd 1982.0 segments (2869936 bytes); observed-rate BDP proxy 1093132 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 cubic run 2: goodput min/median/mean/P95 = 222.768 / 281.348 / 279.725 / 292.683; mean RTT = 75.246; retransmitted segments = 36; matched destination = True.

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Window/RTT context: minimum filtered RTT 30.801 ms; median cwnd 2061.0 segments (2984328 bytes); observed-rate BDP proxy 1076977 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 reno run 2: goodput min/median/mean/P95 = 220.849 / 285.472 / 284.406 / 296.555; mean RTT = 73.469; retransmitted segments = 63; matched destination = True.

Window/RTT context: minimum filtered RTT 31.606 ms; median cwnd 4657.0 segments (6743336 bytes); observed-rate BDP proxy 1123614 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 bbr run 2: goodput min/median/mean/P95 = 197.247 / 282.652 / 277.170 / 295.825; mean RTT = 71.937; retransmitted segments = 17; matched destination = True.

Window/RTT context: minimum filtered RTT 29.064 ms; median cwnd 3533.0 segments (5115784 bytes); observed-rate BDP proxy 1006962 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 reno run 3: goodput min/median/mean/P95 = 217.907 / 283.042 / 280.354 / 297.142; mean RTT = 74.125; retransmitted segments = 36; matched destination = True.

Window/RTT context: minimum filtered RTT 31.095 ms; median cwnd 7854.0 segments (11372592 bytes); observed-rate BDP proxy 1089693 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 bbr run 3: goodput min/median/mean/P95 = 209.804 / 281.981 / 275.889 / 297.117; mean RTT = 59.194; retransmitted segments = 75; matched destination = True.

Window/RTT context: minimum filtered RTT 29.426 ms; median cwnd 2598.0 segments (3761904 bytes); observed-rate BDP proxy 1014780 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

dal.speedtest.clouvider.net:5200 cubic run 3: goodput min/median/mean/P95 = 17.659 / 277.087 /

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268.470 / 292.971; mean RTT = 64.241; retransmitted segments = 67; matched destination = True.

Window/RTT context: minimum filtered RTT 29.714 ms; median cwnd 2006.0 segments (2904688 bytes); observed-rate BDP proxy 997158 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

a110.speedtest.wobcom.de:5201 bbr run 2: goodput min/median/mean/P95 = 13.759 / 173.237 / 163.875 / 178.361; mean RTT = 126.951; retransmitted segments = 3; matched destination = False.

Window/RTT context: minimum filtered RTT 119.372 ms; median cwnd 5741.0 segments (8312968 bytes); observed-rate BDP proxy 2445246 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

a110.speedtest.wobcom.de:5201 cubic run 2: goodput min/median/mean/P95 = 14.702 / 174.769 / 170.379 / 176.238; mean RTT = 127.200; retransmitted segments = 3; matched destination = False.

Window/RTT context: minimum filtered RTT 120.371 ms; median cwnd 2566.0 segments (3715568 bytes); observed-rate BDP proxy 2563566 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 bbr run 1: goodput min/median/mean/P95 = 59.552 / 251.440 / 212.975 / 285.891; mean RTT = 93.484; retransmitted segments = 26; matched destination = False.

Window/RTT context: minimum filtered RTT 25.500 ms; median cwnd 2733.0 segments (3957384 bytes); observed-rate BDP proxy 678865 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 cubic run 1: goodput min/median/mean/P95 = 17.736 / 188.765 / 183.977 / 256.022; mean RTT = 107.207; retransmitted segments = 311; matched destination = False.

Window/RTT context: minimum filtered RTT 25.948 ms; median cwnd 1870.5 segments (2708484 bytes); observed-rate BDP proxy 596719 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

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speedtest.wdc2.us.leaseweb.net:5201 reno run 1: goodput min/median/mean/P95 = 72.866 / 222.657 / 207.479 / 269.479; mean RTT = 102.964; retransmitted segments = 69; matched destination = False.

Window/RTT context: minimum filtered RTT 29.514 ms; median cwnd 9199.0 segments (13320152 bytes); observed-rate BDP proxy 765435 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 cubic run 2: goodput min/median/mean/P95 = 141.245 / 250.684 / 245.309 / 275.132; mean RTT = 81.463; retransmitted segments = 59; matched destination = False.

Window/RTT context: minimum filtered RTT 26.539 ms; median cwnd 2031.0 segments (2940888 bytes); observed-rate BDP proxy 813787 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 bbr run 2: goodput min/median/mean/P95 = 46.729 / 246.086 / 223.961 / 280.504; mean RTT = 79.575; retransmitted segments = 216; matched destination = False.

Window/RTT context: minimum filtered RTT 25.950 ms; median cwnd 2370.0 segments (3431760 bytes); observed-rate BDP proxy 726474 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 reno run 3: goodput min/median/mean/P95 = 201.369 / 269.235 / 263.684 / 286.277; mean RTT = 78.320; retransmitted segments = 6; matched destination = False.

Window/RTT context: minimum filtered RTT 26.971 ms; median cwnd 5847.0 segments (8466456 bytes); observed-rate BDP proxy 888971 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speedtest.wdc2.us.leaseweb.net:5201 bbr run 3: goodput min/median/mean/P95 = 63.063 / 274.409 / 251.208 / 288.431; mean RTT = 85.323; retransmitted segments = 12; matched destination = False.

Window/RTT context: minimum filtered RTT 25.407 ms; median cwnd 2843.0 segments (4116664 bytes); observed-rate BDP proxy 797801 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

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speedtest.wdc2.us.leaseweb.net:5201 cubic run 3: goodput min/median/mean/P95 = 203.022 / 268.101 / 262.218 / 279.950; mean RTT = 70.258; retransmitted segments = 8; matched destination = False.

Window/RTT context: minimum filtered RTT 27.164 ms; median cwnd 1999.0 segments (2894552 bytes); observed-rate BDP proxy 890363 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

nl.speed.vps1.net:5201 cubic run 1: goodput min/median/mean/P95 = 34.791 / 192.631 / 183.786 / 200.219; mean RTT = 118.500; retransmitted segments = 5; matched destination = False.

Window/RTT context: minimum filtered RTT 104.721 ms; median cwnd 5277.0 segments (7641096 bytes); observed-rate BDP proxy 2405788 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

nl.speed.vps1.net:5201 reno run 1: goodput min/median/mean/P95 = 29.533 / 187.833 / 183.824 / 195.101; mean RTT = 117.804; retransmitted segments = 15; matched destination = False.

Window/RTT context: minimum filtered RTT 107.944 ms; median cwnd 5692.0 segments (8242016 bytes); observed-rate BDP proxy 2480309 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

nl.speed.vps1.net:5201 bbr run 1: goodput min/median/mean/P95 = 18.260 / 189.426 / 180.351 / 195.177; mean RTT = 116.421; retransmitted segments = 43; matched destination = False.

Window/RTT context: minimum filtered RTT 107.349 ms; median cwnd 5754.0 segments (8331792 bytes); observed-rate BDP proxy 2420024 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

nl.speed.vps1.net:5201 cubic run 2: goodput min/median/mean/P95 = 29.511 / 190.652 / 185.730 / 194.965; mean RTT = 116.274; retransmitted segments = 3; matched destination = False.

Window/RTT context: minimum filtered RTT 108.469 ms; median cwnd 5954.0 segments (8621392 bytes); observed-rate BDP proxy 2518210 bytes. This proxy uses achieved goodput, so it cannot

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independently identify bottleneck capacity.

nl.speed.vps1.net:5201 cubic run 3: goodput min/median/mean/P95 = 29.523 / 192.030 / 187.404 / 196.124; mean RTT = 116.371; retransmitted segments = 6; matched destination = False.

Window/RTT context: minimum filtered RTT 107.013 ms; median cwnd 5914.0 segments (8563472 bytes); observed-rate BDP proxy 2506805 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 reno run 1: goodput min/median/mean/P95 = 3.588 / 122.907 / 118.041 / 127.958; mean RTT = 185.027; retransmitted segments = 12; matched destination = True.

Window/RTT context: minimum filtered RTT 171.392 ms; median cwnd 5952.0 segments (8618496 bytes); observed-rate BDP proxy 2528875 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 bbr run 1: goodput min/median/mean/P95 = 2.512 / 120.574 / 115.562 / 128.968; mean RTT = 178.686; retransmitted segments = 2; matched destination = True.

Window/RTT context: minimum filtered RTT 173.182 ms; median cwnd 5720.0 segments (8282560 bytes); observed-rate BDP proxy 2501635 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 cubic run 1: goodput min/median/mean/P95 = 3.588 / 122.346 / 120.061 / 127.930; mean RTT = 179.203; retransmitted segments = 17; matched destination = True.

Window/RTT context: minimum filtered RTT 173.757 ms; median cwnd 5942.0 segments (8604016 bytes); observed-rate BDP proxy 2607640 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 bbr run 2: goodput min/median/mean/P95 = 2.465 / 123.697 / 117.634 / 129.107; mean RTT = 176.741; retransmitted segments = 0; matched destination = True.

Window/RTT context: minimum filtered RTT 171.170 ms; median cwnd 5716.0 segments (8276768 bytes);

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observed-rate BDP proxy 2516855 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 cubic run 2: goodput min/median/mean/P95 = 3.586 / 117.321 / 113.307 / 123.802; mean RTT = 187.737; retransmitted segments = 2; matched destination = True.

Window/RTT context: minimum filtered RTT 173.293 ms; median cwnd 2622.0 segments (3796656 bytes); observed-rate BDP proxy 2454387 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 reno run 2: goodput min/median/mean/P95 = 3.590 / 122.629 / 119.492 / 126.308; mean RTT = 180.535; retransmitted segments = 20; matched destination = True.

Window/RTT context: minimum filtered RTT 173.809 ms; median cwnd 5820.0 segments (8427360 bytes); observed-rate BDP proxy 2596090 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 cubic run 3: goodput min/median/mean/P95 = 3.589 / 123.445 / 120.891 / 128.412; mean RTT = 178.770; retransmitted segments = 4; matched destination = True.

Window/RTT context: minimum filtered RTT 170.633 ms; median cwnd 3964.0 segments (5739872 bytes); observed-rate BDP proxy 2578467 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 reno run 3: goodput min/median/mean/P95 = 3.589 / 122.837 / 119.667 / 127.144; mean RTT = 179.453; retransmitted segments = 9; matched destination = True.

Window/RTT context: minimum filtered RTT 173.343 ms; median cwnd 5740.0 segments (8311520 bytes); observed-rate BDP proxy 2592920 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

speed.netfiber.net.il:5201 bbr run 3: goodput min/median/mean/P95 = 2.176 / 90.942 / 86.574 / 120.681; mean RTT = 176.731; retransmitted segments = 396; matched destination = True.

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Window/RTT context: minimum filtered RTT 172.429 ms; median cwnd 4060.0 segments (5878880 bytes); observed-rate BDP proxy 1865930 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 reno run 1: goodput min/median/mean/P95 = 3.607 / 135.962 / 132.816 / 139.885; mean RTT = 162.959; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 154.782 ms; median cwnd 4045.0 segments (5760080 bytes); observed-rate BDP proxy 2569662 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 cubic run 2: goodput min/median/mean/P95 = 7.171 / 135.784 / 132.816 / 139.378; mean RTT = 162.767; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 155.357 ms; median cwnd 4029.0 segments (5737296 bytes); observed-rate BDP proxy 2579196 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 reno run 2: goodput min/median/mean/P95 = 6.435 / 132.785 / 129.360 / 135.174; mean RTT = 166.385; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 159.340 ms; median cwnd 4036.0 segments (5747264 bytes); observed-rate BDP proxy 2576474 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 bbr run 2: goodput min/median/mean/P95 = 2.921 / 127.339 / 120.666 / 131.112; mean RTT = 172.105; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 166.093 ms; median cwnd 5853.0 segments (8334672 bytes); observed-rate BDP proxy 2505220 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 reno run 3: goodput min/median/mean/P95 = 3.254 / 124.793 / 122.265 / 139.106; mean RTT = 176.513; retransmitted segments = 0; matched destination = False.

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Window/RTT context: minimum filtered RTT 153.796 ms; median cwnd 4030.0 segments (5738720 bytes); observed-rate BDP proxy 2350437 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 bbr run 3: goodput min/median/mean/P95 = 2.853 / 126.108 / 120.430 / 131.095; mean RTT = 172.668; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 166.846 ms; median cwnd 5788.0 segments (8242112 bytes); observed-rate BDP proxy 2511609 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

iperf.pendc.com:5201 cubic run 3: goodput min/median/mean/P95 = 3.050 / 131.528 / 110.381 / 135.569; mean RTT = 184.828; retransmitted segments = 0; matched destination = False.

Window/RTT context: minimum filtered RTT 158.601 ms; median cwnd 2049.0 segments (2917776 bytes); observed-rate BDP proxy 2188278 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 cubic run 1: goodput min/median/mean/P95 = 23.359 / 174.920 / 169.732 / 179.385; mean RTT = 127.934; retransmitted segments = 35; matched destination = True.

Window/RTT context: minimum filtered RTT 117.895 ms; median cwnd 3970.0 segments (5748560 bytes); observed-rate BDP proxy 2501320 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 reno run 1: goodput min/median/mean/P95 = 24.358 / 180.366 / 176.219 / 185.585; mean RTT = 121.692; retransmitted segments = 3; matched destination = True.

Window/RTT context: minimum filtered RTT 114.704 ms; median cwnd 4166.0 segments (6032368 bytes); observed-rate BDP proxy 2526598 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 bbr run 1: goodput min/median/mean/P95 = 0.000 / 176.741 / 148.138 / 186.682;

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mean RTT = 192.792; retransmitted segments = 435; matched destination = True.

Window/RTT context: minimum filtered RTT 112.199 ms; median cwnd 5749.0 segments (8324552 bytes); observed-rate BDP proxy 2077564 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 reno run 2: goodput min/median/mean/P95 = 16.486 / 159.570 / 152.885 / 183.652; mean RTT = 141.376; retransmitted segments = 6; matched destination = True.

Window/RTT context: minimum filtered RTT 114.346 ms; median cwnd 3974.0 segments (5754352 bytes); observed-rate BDP proxy 2185195 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 bbr run 2: goodput min/median/mean/P95 = 12.026 / 164.477 / 153.135 / 180.577; mean RTT = 141.953; retransmitted segments = 7; matched destination = True.

Window/RTT context: minimum filtered RTT 109.866 ms; median cwnd 5780.0 segments (8369440 bytes); observed-rate BDP proxy 2102985 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 cubic run 2: goodput min/median/mean/P95 = 12.570 / 156.311 / 143.270 / 170.208; mean RTT = 147.650; retransmitted segments = 0; matched destination = True.

Window/RTT context: minimum filtered RTT 119.790 ms; median cwnd 2030.0 segments (2939440 bytes); observed-rate BDP proxy 2145256 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 bbr run 3: goodput min/median/mean/P95 = 2.244 / 183.417 / 141.152 / 190.987; mean RTT = 117.061; retransmitted segments = 99; matched destination = True.

Window/RTT context: minimum filtered RTT 108.886 ms; median cwnd 4032.0 segments (5838336 bytes); observed-rate BDP proxy 1921160 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

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96.45.41.167:30001 cubic run 3: goodput min/median/mean/P95 = 29.351 / 186.270 / 176.354 / 193.780; mean RTT = 122.158; retransmitted segments = 85; matched destination = True.

Window/RTT context: minimum filtered RTT 109.457 ms; median cwnd 3965.0 segments (5741320 bytes); observed-rate BDP proxy 2412853 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

96.45.41.167:30001 reno run 3: goodput min/median/mean/P95 = 20.127 / 178.491 / 161.471 / 186.671; mean RTT = 147.005; retransmitted segments = 2; matched destination = True.

Window/RTT context: minimum filtered RTT 112.252 ms; median cwnd 3984.0 segments (5768832 bytes); observed-rate BDP proxy 2265662 bytes. This proxy uses achieved goodput, so it cannot independently identify bottleneck capacity.

Interpretation: cwnd limits in-flight segments; its byte budget is cwnd * MSS. Without another bottleneck, the window-limited rate is approximately cwnd * MSS / RTT. The bandwidth-delay product estimates bytes needed in flight at bottleneck bandwidth and base RTT. Extra in-flight bytes can accumulate in queues. RTT growth at similar goodput is consistent with extra queueing; a larger cwnd alone does not establish higher useful throughput.

Reno increases its congestion window additively during congestion avoidance and reduces it after loss. CUBIC probes with a cubic window trajectory and can recover capacity faster on large bandwidth-delay paths. BBR estimates delivery bandwidth and propagation RTT and uses a model with pacing. These are expected mechanisms, not evidence that a particular algorithm won on an unmeasured network regime. The installed BBR version depends on the host kernel.

Retransmissions are a loss proxy, not a direct packet-loss probability. Reordering and spurious retransmissions can contribute. Window reductions may precede a later goodput drop; simultaneous scatter plots do not capture every delayed response. Public-server rate caps, receiver windows, CPU limits, and application limits can hide congestion-control differences.

Limitations: sequential tests share destinations and settings but cannot hold Internet cross-traffic or routes constant. Rotating order reduces systematic order bias without eliminating it. Short runs include startup. Excluding failed candidate groups can bias the sample toward reliable

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servers. Differences of the same magnitude as run variability are not strong evidence of an advantage. Establishing operating regimes causally requires controlled bottleneck bandwidth, RTT, queue size, and loss; public traces alone are insufficient.

AI assistance: we used AI tools to generate and revise substantial portions of the Python client, experiment automation, tests, and plotting/reporting code, and to help refine the controller design, analysis, and explanations.

Per-run goodput summary (Mbit/s); * = incomplete destination group

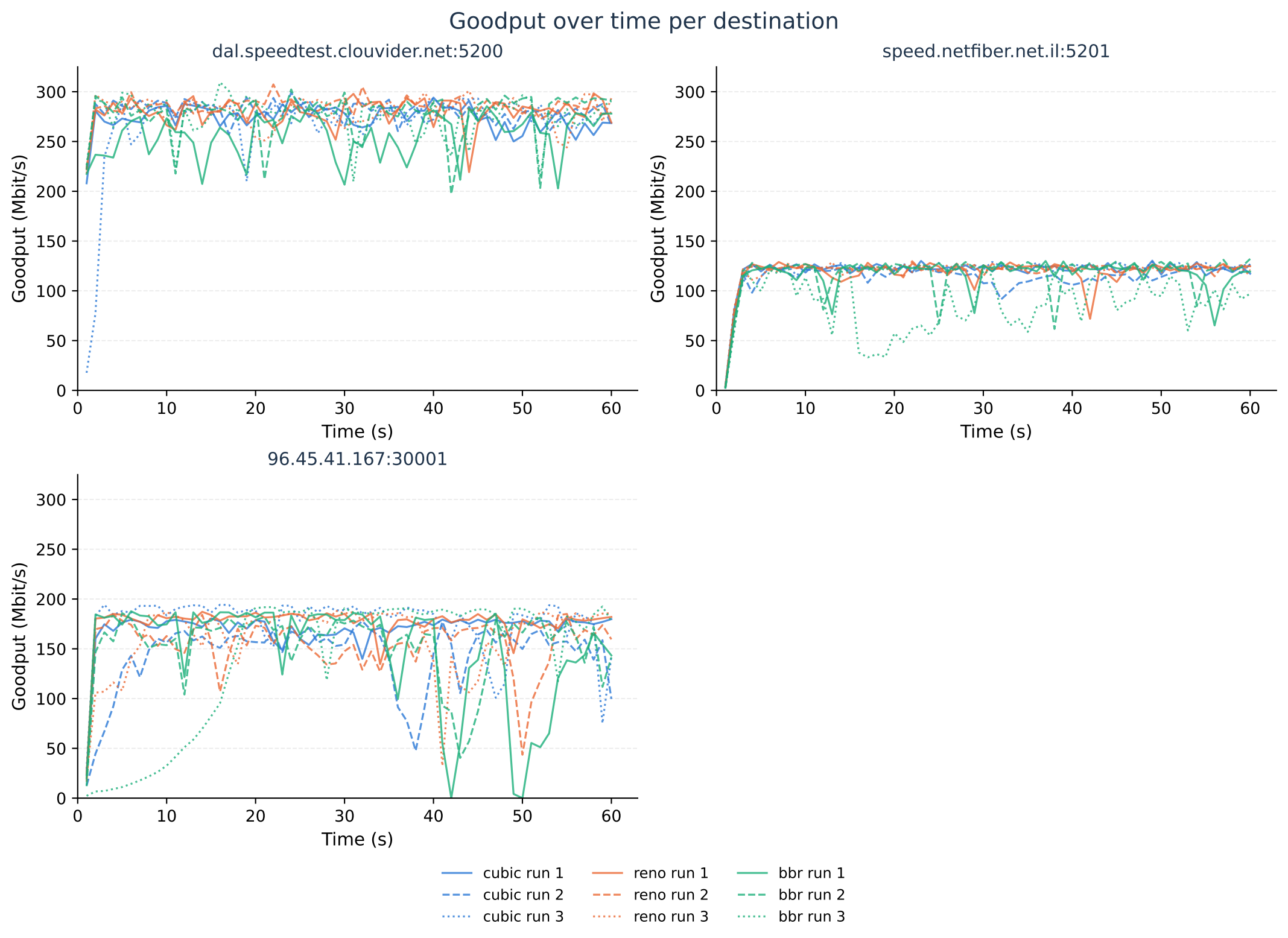
Destination	Algorithm	Run	Min	Median	Mean	P95
dal.speedtest.clouvider.net:5200	bbr	1	202.814	260.731	256.315	281.472
dal.speedtest.clouvider.net:5200	cubic	1	208.026	275.568	273.692	287.345
dal.speedtest.clouvider.net:5200	reno	1	219.239	281.231	279.146	293.892
dal.speedtest.clouvider.net:5200	cubic	2	222.768	281.348	279.725	292.683
dal.speedtest.clouvider.net:5200	reno	2	220.849	285.472	284.406	296.555
dal.speedtest.clouvider.net:5200	bbr	2	197.247	282.652	277.170	295.825
dal.speedtest.clouvider.net:5200	reno	3	217.907	283.042	280.354	297.142
dal.speedtest.clouvider.net:5200	bbr	3	209.804	281.981	275.889	297.117
dal.speedtest.clouvider.net:5200	cubic	3	17.659	277.087	268.470	292.971
a110.speedtest.wobcom.de:5201*	bbr	2	13.759	173.237	163.875	178.361
a110.speedtest.wobcom.de:5201*	cubic	2	14.702	174.769	170.379	176.238
speedtest.wdc2.us.leaseweb.net:5201*	bbr	1	59.552	251.440	212.975	285.891
speedtest.wdc2.us.leaseweb.net:5201*	cubic	1	17.736	188.765	183.977	256.022
speedtest.wdc2.us.leaseweb.net:5201*	reno	1	72.866	222.657	207.479	269.479
speedtest.wdc2.us.leaseweb.net:5201*	cubic	2	141.245	250.684	245.309	275.132
speedtest.wdc2.us.leaseweb.net:5201*	bbr	2	46.729	246.086	223.961	280.504
speedtest.wdc2.us.leaseweb.net:5201*	reno	3	201.369	269.235	263.684	286.277
speedtest.wdc2.us.leaseweb.net:5201*	bbr	3	63.063	274.409	251.208	288.431

Per-run goodput summary (Mbit/s); * = incomplete destination group

Destination	Algorithm	Run	Min	Median	Mean	P95
speedtest.wdc2.us.leaseweb.net:5201*	cubic	3	203.022	268.101	262.218	279.950
nl.speed.vps1.net:5201*	cubic	1	34.791	192.631	183.786	200.219
nl.speed.vps1.net:5201*	reno	1	29.533	187.833	183.824	195.101
nl.speed.vps1.net:5201*	bbr	1	18.260	189.426	180.351	195.177
nl.speed.vps1.net:5201*	cubic	2	29.511	190.652	185.730	194.965
nl.speed.vps1.net:5201*	cubic	3	29.523	192.030	187.404	196.124
speed.netfiber.net.il:5201	reno	1	3.588	122.907	118.041	127.958
speed.netfiber.net.il:5201	bbr	1	2.512	120.574	115.562	128.968
speed.netfiber.net.il:5201	cubic	1	3.588	122.346	120.061	127.930
speed.netfiber.net.il:5201	bbr	2	2.465	123.697	117.634	129.107
speed.netfiber.net.il:5201	cubic	2	3.586	117.321	113.307	123.802
speed.netfiber.net.il:5201	reno	2	3.590	122.629	119.492	126.308
speed.netfiber.net.il:5201	cubic	3	3.589	123.445	120.891	128.412
speed.netfiber.net.il:5201	reno	3	3.589	122.837	119.667	127.144
speed.netfiber.net.il:5201	bbr	3	2.176	90.942	86.574	120.681
iperf.pendc.com:5201*	reno	1	3.607	135.962	132.816	139.885
iperf.pendc.com:5201*	cubic	2	7.171	135.784	132.816	139.378
iperf.pendc.com:5201*	reno	2	6.435	132.785	129.360	135.174

Per-run goodput summary (Mbit/s); * = incomplete destination group

Destination	Algorithm	Run	Min	Median	Mean	P95
iperf.pendc.com:5201*	bbr	2	2.921	127.339	120.666	131.112
iperf.pendc.com:5201*	reno	3	3.254	124.793	122.265	139.106
iperf.pendc.com:5201*	bbr	3	2.853	126.108	120.430	131.095
iperf.pendc.com:5201*	cubic	3	3.050	131.528	110.381	135.569
96.45.41.167:30001	cubic	1	23.359	174.920	169.732	179.385
96.45.41.167:30001	reno	1	24.358	180.366	176.219	185.585
96.45.41.167:30001	bbr	1	0.000	176.741	148.138	186.682
96.45.41.167:30001	reno	2	16.486	159.570	152.885	183.652
96.45.41.167:30001	bbr	2	12.026	164.477	153.135	180.577
96.45.41.167:30001	cubic	2	12.570	156.311	143.270	170.208
96.45.41.167:30001	bbr	3	2.244	183.417	141.152	190.987
96.45.41.167:30001	cubic	3	29.351	186.270	176.354	193.780
96.45.41.167:30001	reno	3	20.127	178.491	161.471	186.671


Goodput over time per destination
dal.speedtest.clouvider.net:5200
speed.netfiber.net.il:5201
96.45.41.167:30001
Goodput (Mbit/s)
Time (s)
0
50
100
150
200
250
300
10
20
30
40
50
60
cubic run 1
cubic run 2
cubic run 3
reno run 1
reno run 2
reno run 3
bbr run 1
bbr run 2
bbr run 3

dal.speedtest.clouvider.net:5200 | time series; all repetitions; common scales per metric

cubic | reno | bbr

Congestion window (segments)

RTT (ms)

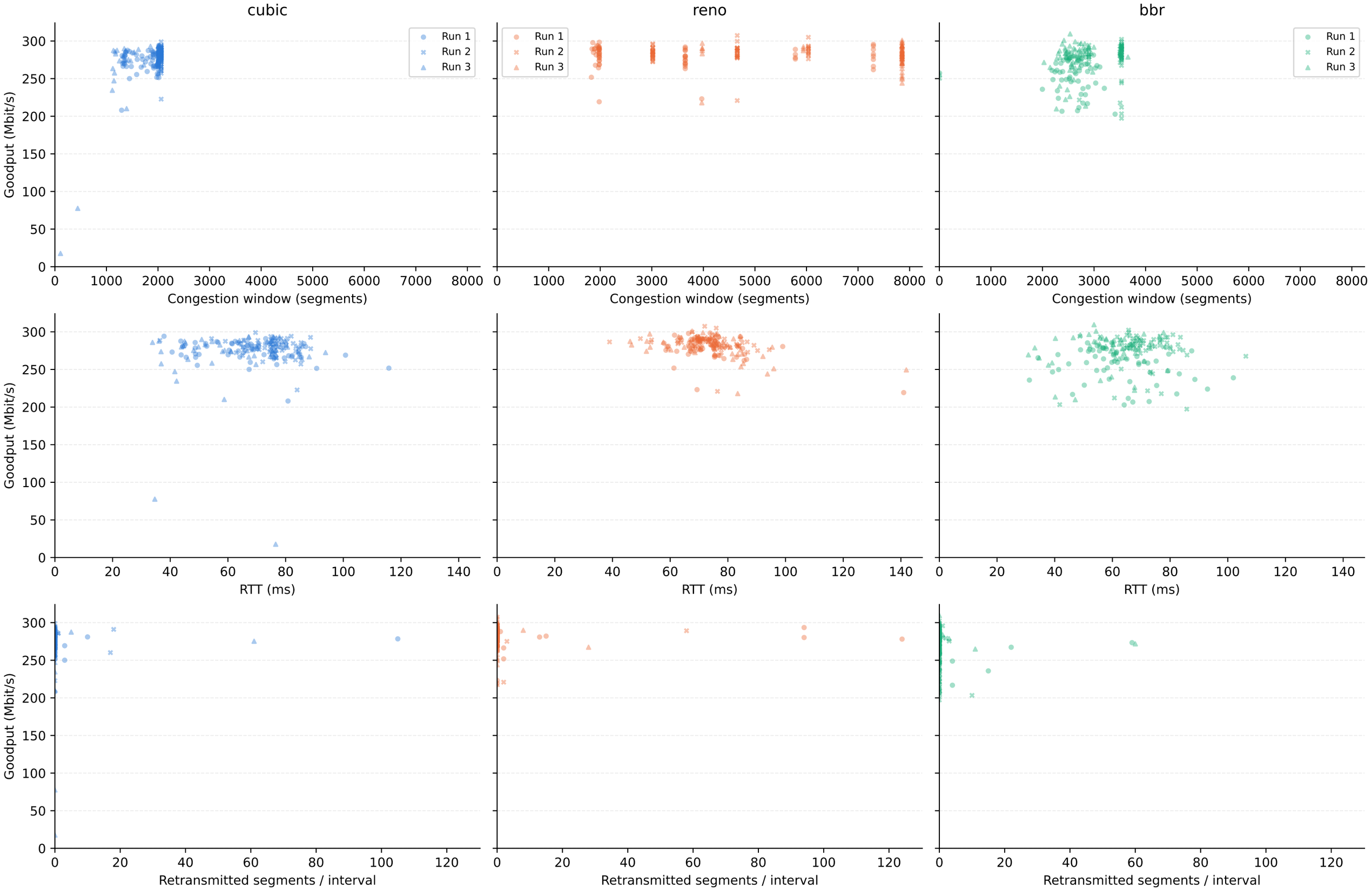
Retransmitted segments / interval

Goodput (Mbit/s)

Time (s)

Run 1
Run 2
Run 3

dal.speedtest.clouvider.net:5200 | scatter; all repetitions; common scales per metric



Controller design reference (Q3_DESIGN.md)

Q3(b): RTT, goodput, and loss based window control

This design was developed for Assignment 2, Question 3(b), and implemented as `a3_cc` in Assignment 3. The Assignment 2 measurements motivated its rules; the Assignment 3 experiments compare it with CUBIC, Reno, and BBR.

AI tools helped develop and simplify this design and refine its explanation.

Idea

Use three signals with clear roles:

- Loss: reduce the sending window through Linux's recovery mechanism.
- RTT: grow cautiously when delay is low; hold when delay is moderately higher.
- Goodput: when RTT is high, reduce the window if delivery is no longer improving.

The measurements motivate this combination. Dallas showed that a much larger

Controller design reference (Q3_DESIGN.md)

window did not necessarily increase throughput. Netfiber had a slow run with many retransmissions. The IP destination had a brief stall with a large RTT increase. These observations motivated the rules; they do not establish an advantage over the baseline algorithms.

State and measurements

Start with an initial window of 10 segments, subject to the TCP stack's limits. Start in slow start, with the slow-start threshold initially unlimited. Keep the lowest valid RTT measured since connecting and the previous eligible interval's goodput. Goodput is newly acknowledged application bytes divided by actual elapsed time, expressed in bytes/s. Window decisions use the socket's smoothed RTT; RTT comparisons use consistent time units.

Make one normal window decision per completed RTT with new ACK progress. During recovery, leave sending and retransmissions to Linux. Also skip normal decisions when the sender has no data ready or the receiver's window limits sending. Clear

Controller design reference (Q3_DESIGN.md)

the previous goodput when skipping an interval or restarting after recovery, so a

pause is not compared with a busy interval.

Pseudocode

```text
State: slow_start = true; previous_goodput = unset
On a new loss-recovery episode:
    request ssthresh = max(2, floor(cwnd / 2))
    let Linux apply recovery once using that threshold
    slow_start = false; clear previous_goodput and interval accounting
On a retransmission timeout:
    let Linux apply its normal timeout window and recovery
    slow_start = true; clear previous_goodput and interval accounting
Once per RTT with new ACK progress, outside recovery:
    update minimum_RTT and measure goodput
    if application limited or receiver-window limited:
```

Controller design reference (Q3_DESIGN.md)

```
        clear previous_goodput; finish interval
    if slow_start and cwnd < ssthresh and RTT <= 1.10 * minimum_RTT:
        cwnd += min(newly_acked_segments, cwnd)
    else:
        slow_start = false
        if RTT > 1.25 * minimum_RTT and previous_goodput is valid
           and goodput <= previous_goodput:
            cwnd = max(2, floor(0.90 * cwnd)); ssthresh = cwnd
        else if RTT <= 1.10 * minimum_RTT:
            cwnd += 1
        else:
            hold cwnd
    obey the kernel window limit
    previous_goodput = goodput; finish interval
```

Linux handles timeout recovery and can reduce the window to one segment. The

Controller design reference (Q3_DESIGN.md)

controller supplies the loss threshold and resets interval history without applying a second recovery reduction. ACK accounting counts newly acknowledged segments rather than ACK messages.

Choices and limitations

The 10% and 25% delay thresholds leave a range where the sender holds its window. The 10% delay reduction is milder than the loss response. These thresholds are heuristics, not tuned settings. Congestion avoidance increases the window by at most one segment per eligible RTT.

The delay and throughput decisions need only a minimum RTT and the previous eligible interval's goodput. The minimum cannot adapt upward after a route change. One-interval goodput comparisons can be noisy, and a single low RTT can make the baseline too optimistic.

With similar RTTs and a shared bottleneck, additive growth and loss reductions

Controller design reference (Q3_DESIGN.md)

may help flows share bandwidth. Flows with shorter RTTs update more often, and this delay-sensitive controller may yield to Reno or CUBIC. Repeated increases and reductions may produce oscillation. The saved single-flow experiments do not establish fairness or convergence between competing flows.